Projeto No. 2

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POLYMATHLAB

Trauma Release Exercises (TRE)
Foldable Chair

March 2026

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Chapter 1

Introduction

Trauma Release Exercises (TRE) is a practice that aims to activate the inherent tremor mechanism present in all mammals. The purpose of the neurogenic tremors is to release stuck sympathetic energy in the nervous system, thus releasing trauma and deep tension patterns in the body. The tremors are activated via a series of exercises intended to fatigue muscles and can be performed standing, lying down or sitting.

While resting between sets of leg presses at a 45º inclination at my local gym, I noticed my legs and hips would shake substantially more on the equipment chair than when practicing TRE laying down on a yoga mat. Projeto No. 2 is a foldable chair inspired by that leg press 45º at the gym so I can practice TRE on it.

The chair is made of both new and scrap pieces of wood that my friend and neighbor, Tácito Vieiro had in his garage. Tácito had the tools, workspace, knowledge, and experience with carpentry and woodworking necessary to complete the project.

The chair is composed of two 2 x 4 rafters that act as the base of the chair, two adjustable sitting rafters and two adjustable backrest rafters. Both the seat and backrest are made of scrap plywood.

The seat angle can be changed between 20º, 30º and 45º while the backrest angle can be adjusted between 25º, 35º and 45º.

Chapter 2

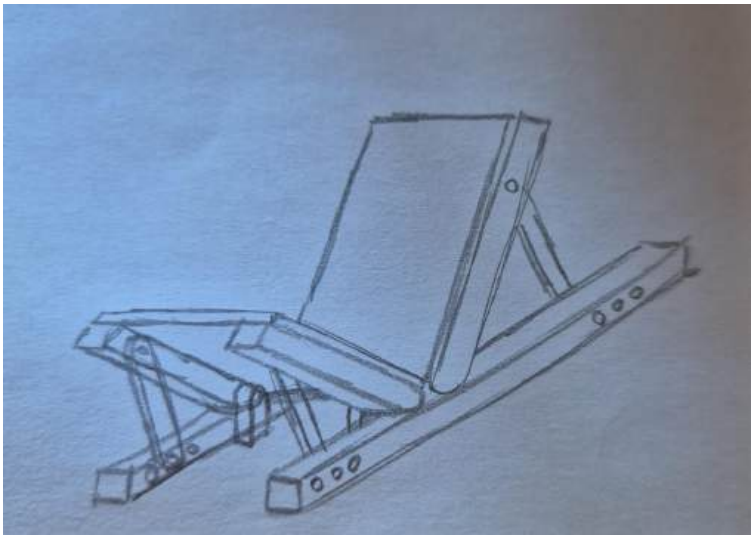
Ideation

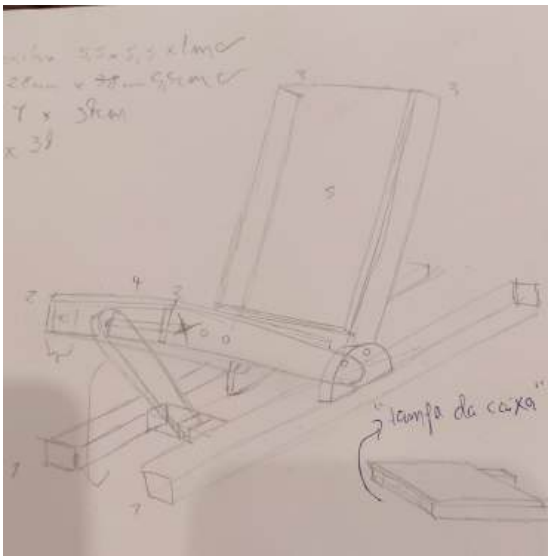
The main inspiration for the TRE chair was the 45º leg press equipment chair at my local gym.

Figure 2.1: 45º leg press.

After taking the measurements for the length, width, height, and angles of the reference

chair, Tácito and I began the ideation phase. We considered a few designs before agreeing on a system of adjustable angles for the seat and backrest, with removable supports to select the desired combination of angles.





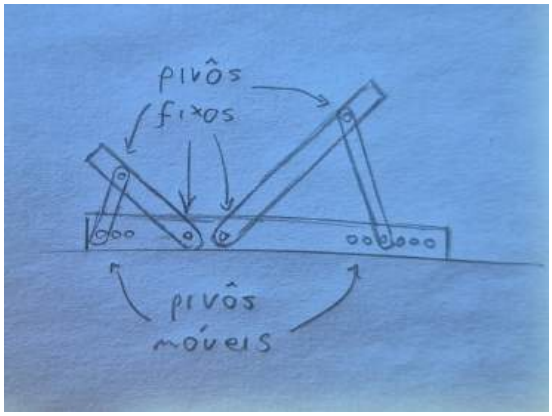
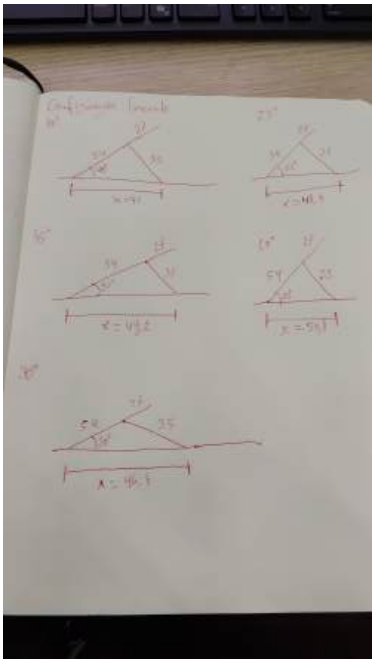
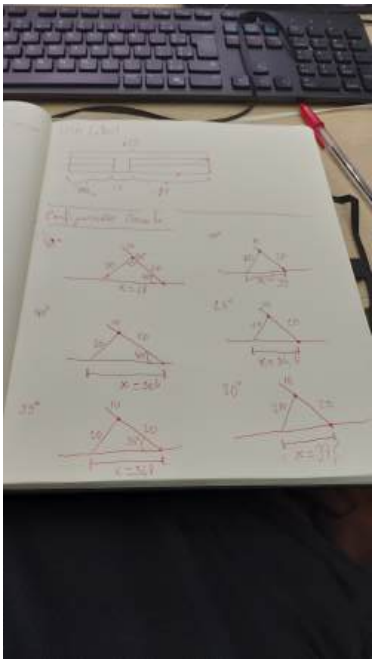


Figure 2.2: Initial drawings.

Initially, many calculations were made, but for simplification purposes, we chose a system where the supporting beams would always be at a 90º angle relative to the base.



(a) Backrest angle calculations.



(b) Seat angle calculations.

Figure 2.3: Angle calculations.

With that, we manually marked where along the length of the base rafters the beams should be for the desired angles of the seat and backrest.

Figure 2.4: Manual angle calculations.

Chapter 3

Build

We began by cutting 2 x 4s to the correct lengths. The base rafter is 75 cm long, the seat is 39.5 cm long and the backrest is 81 cm long.

Figure 3.1: 2 x 4 rafters cut to length.

5 cm-wide hinges were used to attach the supporting rafters to the base rafters and to provide angle adjustment.

Figure 3.2: Hinges installation.

Because the screws were parallel to the grain, one issue we encountered during hinge installation was that they kept coming loose whenever the beams were moved. In order to solve this, we cut and inserted a small squared piece to every hinge attachment in an angle where the screws would go against the grain to provide additional strength. These were later cut with a Japanese saw to stay flush with the adjustable rafters and so that the seat and backrest could be installed flat.

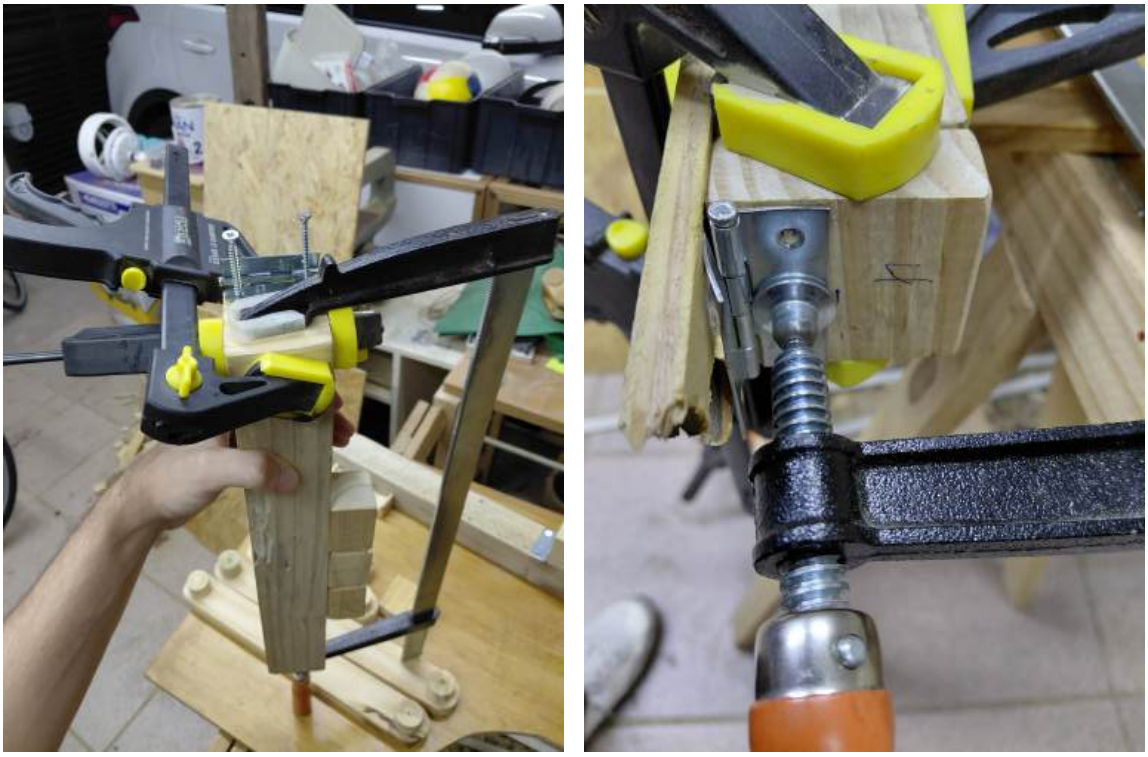


Figure 3.3: Support pieces of wood for screwing against the grain.

With the base and supporting beams sanded and finished, next, came the supporting cups. These hold the supporting beams in specific locations along the base rafters and allow for changing between seat and backrest angles. There are a total of 24 cups positioned on both the base and supporting rafters, and allow for a total of 9 combinations of angles.

We clamped together two 2x2 pieces of wood - which halved the time to cut them afterwards - and measured 3.5 cm-diameter circles with each half of the circles positioned on the edges of the wood so we could have half-circled cups.

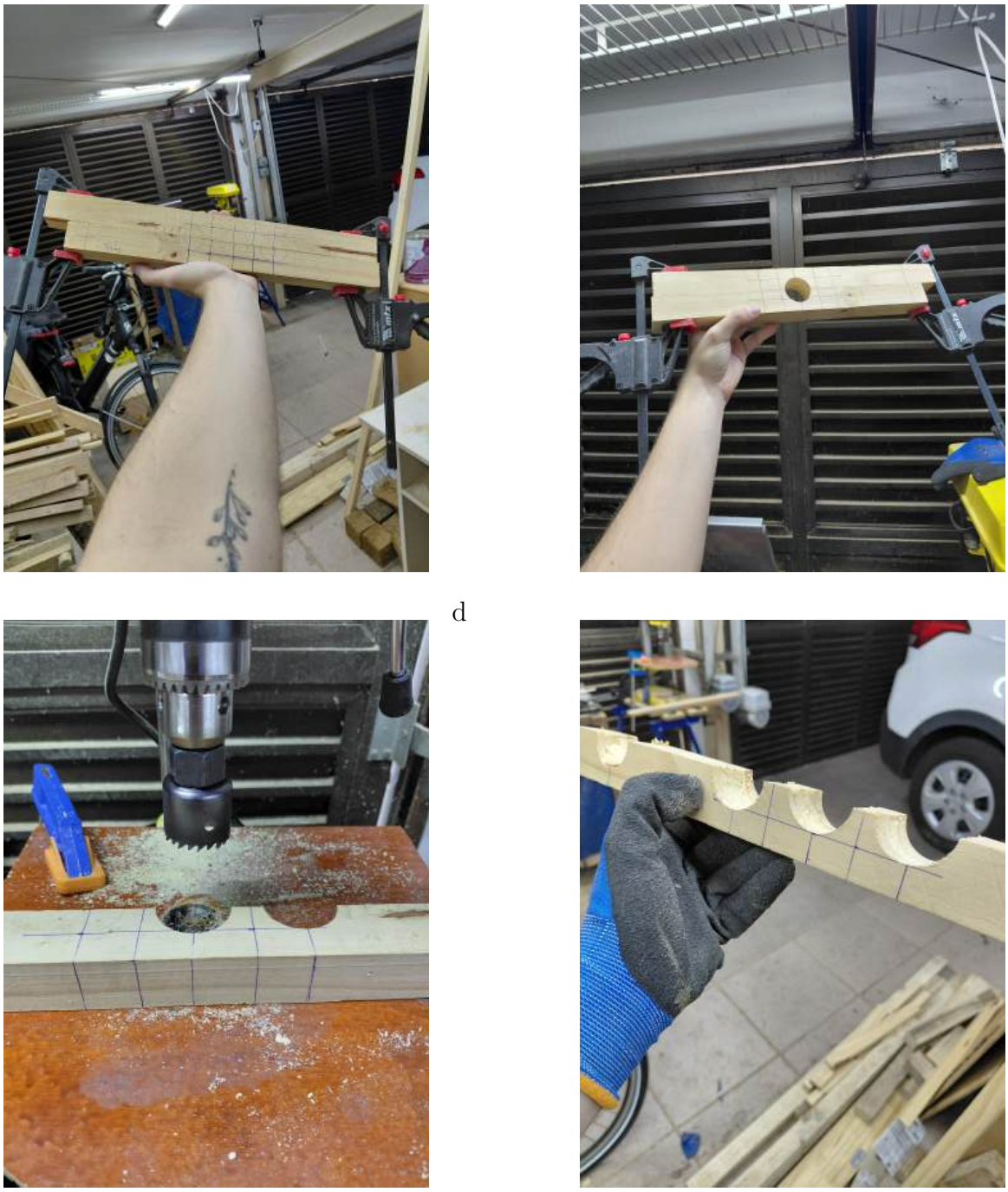


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Figure 3.4: Making of support cups.

The cups were then cut into individual pieces and screwed to the base and supporting rafters in the marked locations for correct angle selection.

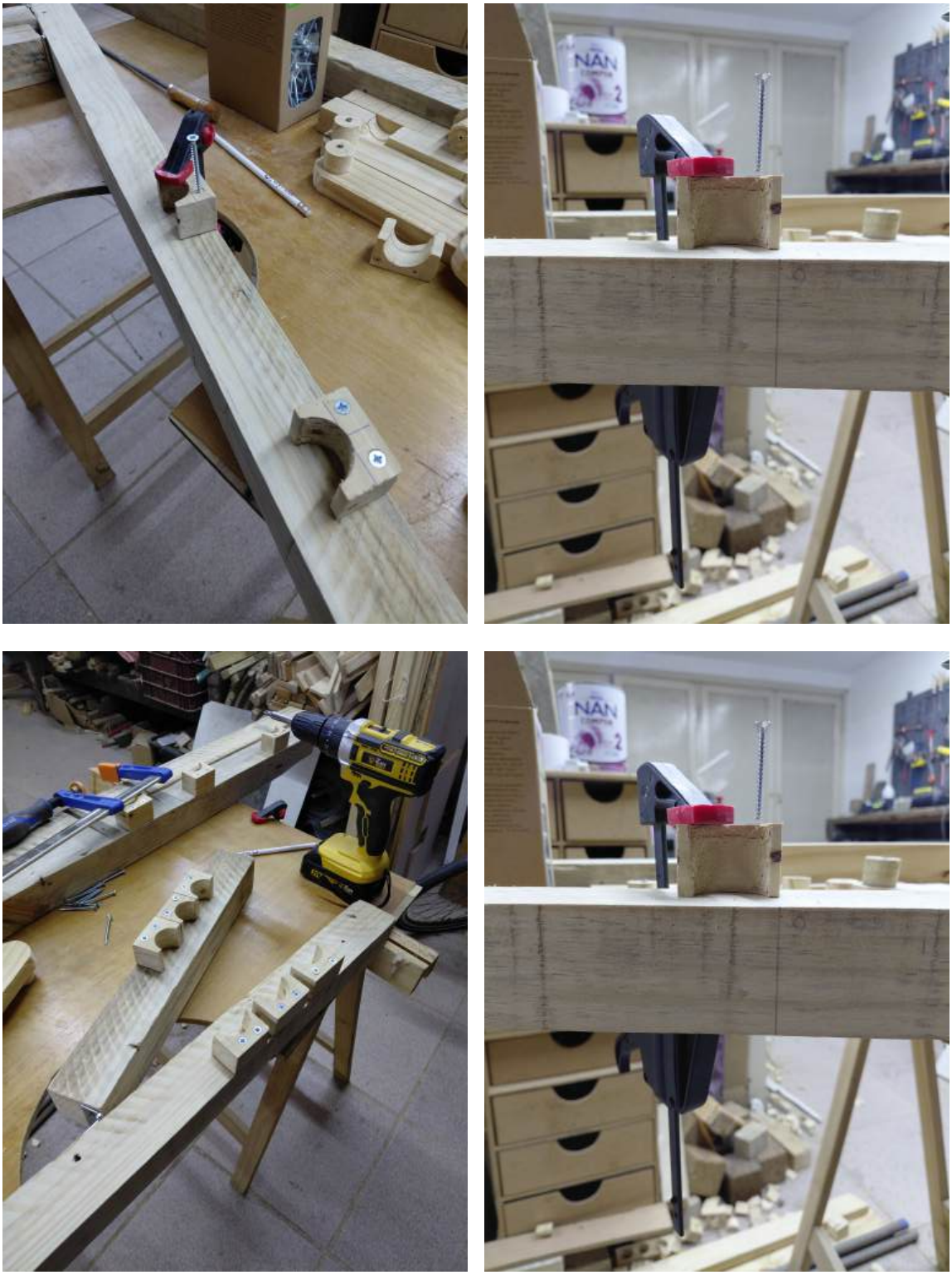

Figure 3.5: Installation of support cups.

Next, we made the supporting beams. They are removable pieces that allow for the choice of seat and backrest positions and support the weight. We cut four pieces to the necessary lengths according to the calculations with a table saw and screwed cylindrical pieces of wood to each end. These act as contact points between the cups.

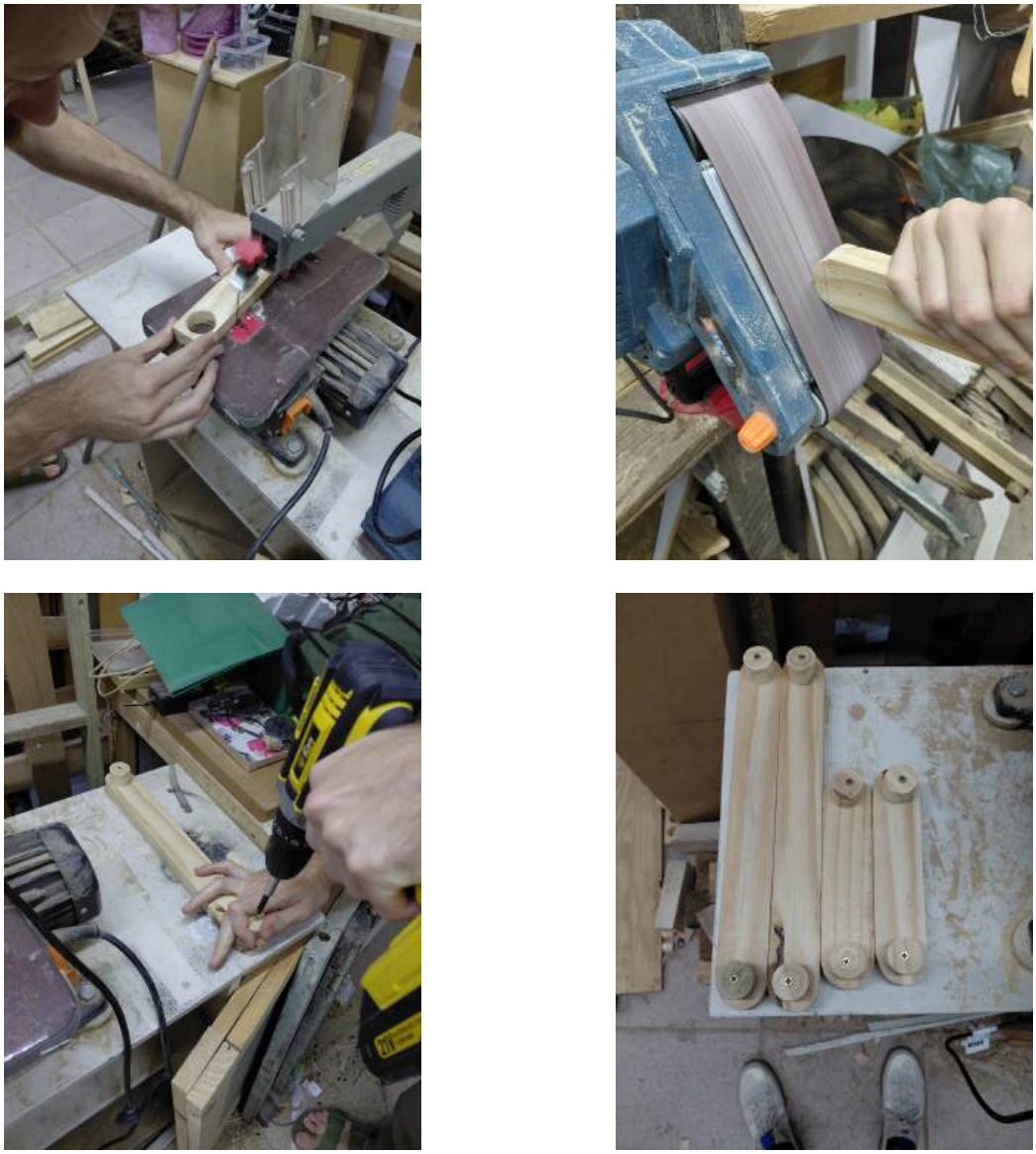

Figure 3.6: Making of supporting beams.

The separate pieces were connected with a piece of wood with the correct width between the left and right supporting cups.

(a) Connecting the large support beam.

(b) Connecting the small support beam.

Figure 3.7: Connecting the supporting beams together.

With all structural pieces installed and secured, we finally screwed two pieces of oriented strand boards: one for the seat and one for the backrest. The seat is 41 x 39.5 cm and the backrest is 81 x 39.5 cm.

(a) Installing seat.

(b) Seat and Backrest installed.

Figure 3.8: Installation of seat and backrest.

Chapter 4

Assembly and Final Product

The complete foldable chair has latches that secure the movable parts and a handle to make it portable.

(a) Internal structure angled view.

(b) Internal structure front view.

(c) Bottom view.

(d) Top view.

Figure 4.1: Final assembly.